

INTERNSHIP TASK REPORT

Domain:	Artificial Intelligence & Machine Learning
Organization:	QSkill
Intern Name:	Nitu
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Introduction

This report outlines the successful completion of the machine learning tasks assigned during the QSkill AI/ML Internship. As per the guidelines, two primary projects were selected, developed, and evaluated using Python and industry-standard machine learning libraries:

- **Task 1: Iris Flower Classification** (Supervised Learning - Classification)
- **Task 2: House Price Prediction** (Supervised Learning - Regression)

Both tasks were implemented and validated inside separate Jupyter Notebooks via Google Colab environments.

Task 1: Iris Flower Classification

1. Objective

The primary goal of this project is to classify iris flowers into three distinct species (*Setosa*, *Versicolor*, and *Virginica*) based on four physical measurements: sepal length, sepal width, petal length, and petal width.

2. Methodology & Steps

- **Environment Setup:** Imported core data analysis and visualization libraries including NumPy, Pandas, Matplotlib, and Seaborn.
- **Dataset:** Loaded the classic Iris dataset directly from `sklearn.datasets`.
- **Data Exploration & Visualization:** Converted raw arrays into a structured Pandas DataFrame and generated pairplots to visually identify linear clusters and boundaries between classes.
- **Data Splitting:** Divided the dataset using an 80% training ratio to build the model weights and a 20% testing ratio for evaluation.

- **Model Training:** Implemented and trained a **Logistic Regression** classifier with max iterations set to 200 for complete convergence.

Task 1 Final Metrics

Model Accuracy: 100.00%

Precision / Recall / F1-Score: 1.00 across all classes (Setosa, Versicolor, Virginica).

Task 2: House Price Prediction

1. Objective

The objective of this project is to build a predictive regression model that estimates California housing prices based on structural and demographic attributes like median income, house age, average rooms, and geographic location.

2. Methodology & Steps

- **Dataset:** Utilized the comprehensive **California Housing Dataset** fetched via `sklearn.datasets`.
- **Data Exploration:** Configured features into a structured format and plotted price distribution histograms to analyze data density.
- **Data Splitting:** Partitioned features and targets using an 80:20 split ratio for training and validation tracking.
- **Model Training:** Trained a mathematical **Linear Regression** model to map continuous inputs against house pricing targets.

Task 2 Final Metrics

Mean Squared Error (MSE): ~0.5556 (Low average squared error variance)

R-squared (R2) Score: ~0.5758 (Explains approximately 57.5% of variance correctly)

Conclusion & Skills Gained

By completing these two foundational machine learning workflows, a strong baseline in modern data science paradigms has been established:

- **Exploratory Data Analysis (EDA):** Gained practical proficiency with data transformation via Pandas and pattern plotting via Seaborn.
- **Supervised Learning Pipelines:** Acquired hands-on experience avoiding data leakage through explicit Train-Test splits.

- **Algorithm Application:** Successfully deployed separate frameworks for categorical classification (Logistic Regression) and continuous estimation (Linear Regression).
- **Statistical Diagnostics:** Learned to interpret accuracy reports, precision-recall indices, MSE errors, and variance scores.